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MILITARY SPECIFICATION

PLANERS, METAL CUTTING, OPENSIDE, HYDRAULIC, SHAPER-PLANER TYPE

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers openside, hydraulically operated planer machines commercially known as shaper planers. They are of size and types suitable for accurate machining of workpieces beyond the capacity of conventional shapers.

1.2 Classification. The planers covered by this specification shall be of the following types and sizes. The type and size supplied shall be as specified (see 6.2.1).

TYPE I - SHORT STROKE

SIZE	TABLE WIDTH	TABLE STROKE
	INCHES (MM)	FEET (MM)
1	20 (508)	3 (914)
2	20 (508)	6 (1829)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, Memphis, Tennessee 38114, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the document or by letter.

FSC 3418

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TYPE II - MEDIUM STROKE

SIZE	TABLE WIDTH		TABLE STROKE	
	Inches	(MM)	Feet	(MM)
1	30	(762)	8	(2438)
2	30	(762)	14	(4267)
3	30	(762)	20	(6096)
4	36	(914)	8	(2438)
5	36	(914)	14	(4267)
6	36	(914)	20	(6096)
7	42	(1067)	8	(2438)
8	42	(1067)	14	(4267)
9	42	(1067)	20	(6096)

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2.1), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the Contracting Officer.)

2.1.2 Other Government documents, drawing and publications. The following other Government documents form a part of this specification to the extent specified herein.

U. S. DEPARTMENT OF LABOR

OSHA - Safety and Health Standards (29 CFR 1910), General Industry

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(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)

ASTM-E-380-79 - Metric Practice

(Application for copies should be addressed to the American Society for Testing and Materials 1916 Race St. Philadelphia, PA 19130)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NFPA NO. 79 - Electrical Standards For Metalworking Machine Tools.

B5.1 T-slots, their bolts, nuts, and tongues.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2.1), the contractor shall furnish one complete planer for first article inspection and approval (see 4.3 and 6.3).

* 3.2 Design. The machine shall be new and of the manufacturers' current design which meet the requirements of this specification. The design shall be such as to provide convenient, efficient and safe operation. The machine shall include all components, parts and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage or distortion shall be accessible for adjustment, replacement and repair.

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* 3.2.1 Measurement systems. Unless otherwise specified, either the U. S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in the U. S. Customary System of Units (US). These measurements may be converted to the International System of Units (SI) through the use of the conversion factors and methods specified in ASTM-E-380-79.

* 3.2.1.1 Measuring and indication device calibrations. When specified (see 6.2.1), measuring and indicating devices such as scales, stops, dial indicators, pressure gauges, temperature indicators, and all other similar devices shall be graduated in the specified system (US or SI) of measurements. Regardless of the measurement system used all measuring and indicating devices on the machine shall be graduated in the same system.

* 3.2.1.2 Dual calibrations. When specified (see 6.2.1), measuring and indicating devices shall be graduated in both the US and SI System of Measurements. When a dual US and SI system is furnished, both feed dials shall have independent zero adjustments and both dials shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

* 3.2.2 Reclaimed materials. The machine shall contain reclaimed materials to the maximum extent possible without jeopardizing its intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which restores them to the same chemical composition and physical properties as the materials originally selected for use on the machine. Reclaimed materials shall include iron, steel, copper, brass, aluminum, fiber products, plastics and elastomers that have been collected from discarded solid, liquid, semi-solid, or gaseous waste such as garbage, refuse or sludge.

* 3.2.3 Controls. All operating controls shall be located convenient to the operator at his normal work station. Type II machines shall have a pendant control providing push-button control of all movements except the table.

* 3.2.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the the require-

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ments of "OSHA Safety and Health Standards (29 CFR 1910), "General Industry" that are applicable to the machine itself. Additional safety and health requirements shall be as specified (see 6.2.1).

* 3.2.5 Lubrication. Both type machines shall be equipped with an automatic pressure lubrication system for lubricating the table ways. Type II machine shall also be provided with a central lubrication system and one shot pressure feed to lubricate the feed mechanism, traverse mechanism and other moving parts that require lubrication. Type I machine shall be provided with means to lubricate all moving parts that require lubrication. Each recirculating system shall include a filter which is cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid level. All oil holes, grease fittings and filler caps shall be accessible.

3.2.6 Interchangeability. To provide for replacements of worn or damaged parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine.

3.3 Construction. The machine shall be constructed of parts which are new, without defects, and free of repair. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without permanent distortion.

* 3.3.1 Castings and forgings. All castings and forgings shall be free of scale and mismatching. No process such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

* 3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Those subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

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* 3.3.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice, provided that all surfaces to be painted shall be cleaned of all foreign matter detrimental to painting, and at least one coat of primer and one coat of finish are applied.

* 3.3.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Dials. All dials shall be graduated in the inch system. Graduations indicating stock removal or tool movement shall be in increments of not more than 0.001 inch. Dial diameters shall be such that graduations may be easily read from the normal operating positions. Dials shall be permanently and legibly engraved or etched, and shall have a non-glare finish.

* 3.3.8 Instruction marking and indicating plates. All words on instruction and on indicating plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. When these methods are not practicable, the instructions shall be applied directly on the item by environmentally protected decalcomania transfer, metal wrap-around tag, stencil, silk screen, or by any other method suitable for the use intended which shall be as permanent as the normal life expectancy of the item on or to which it is applied and capable of withstanding the environmental tests and cleaning procedures specified for the item.

* 3.3.9 Gears. All gears and pinions shall be constructed from a material suitable for the intended purpose. All gears and pinions shall be machined and heat treated by a process that will impart the hardness, smoothness and toughness that will enable the gear train to transmit full rated torque of the drive motor without gear damage, failure, or premature deterioration and wear. All gears and pinions shall be manufactured to definite established standards to provide for interchangeability and parts replacement.

3.4 COMPONENTS

3.4.1 Bed. The bed shall support the worktable and column and shall contain the table drive mechanism. The bed shall have a bedway conforming to 3.4.3. The length of the bed and bedways shall be such that the worktable is fully supported by the bedways without overhang with the table in any position within its rated stroke. The bed shall have access openings of adequate size to permit servicing of all components contained within the bed casting. An adequate number of leveling screws and hold-down bolts shall be provided to enable the machine to be properly installed and leveled.

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* 3.4.2 Column. The column shall be aligned, and securely bolted to the base in such a manner that the heaviest cuts within the capacity of the machine shall be taken without causing any looseness between these members. The column shall have vertical ways for supporting the crossrail and the side toolhead. The ways may be either cast integrally with the column casting, or individual way castings aligned and securely attached to the column. The ways shall be of adequate length to allow the crossrail and side toolhead to travel vertically within the ranges specified in table I.

3.4.3 Ways and way slides. All mating ways and way slides shall be machined, ground, handscraped, and fitted in accordance with the manufacturer's standard commercial practice. The accuracy of machining, grinding, hand-scraping and fitting shall be such that the machine will meet all the alignment and performance requirements stated herein. All mating ways and way slides with the exception of the table, shall have full length, one-piece gibs to provide alignment adjustment, and to compensate for wear of the way or slide. Effective way wipers or guards shall be provided on the table and bed ways to prevent entrance of chips, dirt or other extraneous material. Unless otherwise specified, the table bedways shall be either of the metallic or non-metallic type. When only one type is acceptable, the particular type required shall be as specified (see 6.2.1).

3.4.4 Crossrail. Unless otherwise specified (see 6.2.1), the crossrail vertical travel shall be not less than the dimension shown as "standard" in table I. When specified (see 6.2.1), the vertical travel shall be not less than the "optional" vertical travel distances specified in table I. The crossrail and its sliding attachment to the column ways shall have sufficient rigidity to withstand maximum cutting forces without permanent deflection or twist. The crossrail shall have a way for supporting the crossrail toolhead, and shall be of adequate length to allow the toolhead to travel to the distances from the column specified in table I. Unless otherwise specified (see 6.2.1), the crossrail elevating mechanism shall be manually operated on Type I machines, and shall be power operated on Type II machines.

3.4.4.1 Crossrail toolhead. Unless otherwise specified (see 6.2.1), the machine shall have one crossrail toolhead traveling horizontally on the crossrail. The toolhead slide and its mating way shall conform to 3.4.3. The toolhead shall swivel on its saddle through an arc of 180 degrees and shall have a dial conforming to 3.3.7 to indicate the angle to which the head is swiveled. Means shall be provided to securely clamp the toolhead to the saddle at any desired swivel angle, and for securely clamping the saddle to the crossrail. The toolhead shall have a hinged clapper box that swivels not less than 15 degrees to the right and left of the toolslide axis, and shall have means for securely clamping the clapper box in any position within this range. The clapper box shall have provisions for securely holding a cutting tool

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of the size specified in table I. Toolhead shall have means to permit compensating for wear.

3.4.4.1.1 Crossrail toolhead feed system. The crossrail toolhead shall have either a hydraulic or a combination hydraulic-mechanical feed system providing an infinite number of feed rates in both horizontal and both vertical directions. The power feed rate range and total tool travel shall conform to the requirements of table I. The vertical power feed mechanism shall be capable of feeding the tool vertically or in any swivel angle within its specified adjustment range. Means shall also be provided for manually feeding the toolhead in both vertical directions and both horizontal directions. The manual feed screws shall have control wheels or crank handles of convenient size and shall have dials or collars conforming to 3.3.7 to indicate tool movement.

* 3.4.4.1.2 Crossrail toolhead rapid traverse system. Type II machine shall have a power rapid traverse system that will traverse the crossrail toolhead in both horizontal directions at a maximum rate of not less than 80 inches per minute (IPM). When specified (see 6.2.1), Type I machines shall also have a rapid traverse system with capacity of not less than 45 IPM.

* 3.4.5 Siderail toolhead. Unless otherwise specified (see 6.2.1), Type I machine shall have a siderail toolhead traveling vertically on the column ways and Type II machine shall have a siderail traveling vertical on the siderail ways. The toolhead slide and way shall conform to the requirements of 3.4.2 and 3.4.3. The toolhead shall swivel on its saddle through an arc of 180 degrees and shall have a dial conforming to 3.3.7 to indicate the angle to which the toolslide is swiveled. Type I shall have a graduated plate, graduated 30 degrees each side of center. Means shall be provided to securely clamp the toolhead to the saddle at any desired swivel angle and for securely clamping the saddle to the column ways. The toolhead shall have a hinged clapper box that swivels not less than 15 degrees above and below the toolslide axis and shall have means for securely clamping the clapper box in any position within its swivel range. The clapper box shall have provisions for securely holding a cutting tool of the size specified in table I.

3.4.5.1 Siderail toolhead feed system. The siderail toolhead shall have either a hydraulic or combination hydraulic-mechanical feed system providing an infinite number of power feed rates in both vertical directions. The power feed rate range and total tool travel, horizontally and vertically, shall conform to table I. A manual feed system that will feed the toolhead in both vertical directions and both horizontal directions shall also be provided. Manual feed screws shall have control wheels or crank handles of convenient size, and shall have graduated dials conforming to 3.3.7 to indicate tool movement. Means shall be provided for securely clamping the toolslide to the saddle

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after adjustment of depth of cut, and for securely clamping the toolhead saddle to the column way.

* 3.4.5.2 Sidehead rapid elevation system. Type II machines shall have a rapid traverse system capable of elevating or lowering the sidehead at a rate not less than 80 IPM. When specified (see 6.2.1), Type I machines shall also have a power rapid elevation system with capacity of not less than 45 IPM.

3.4.6 Safety stops. Means shall be provided to prevent feeding or rapid traversing the crossrail, crossrail toolhead, or side toolhead off the ends of their ways or into collision with adjacent machine parts or structures. Safety devices shall also be provided to prevent damage to the feed or rapid traverse mechanicsms if the moving part is inadvertently fed or rapid traversed into the workpiece or the safety stops.

3.4.7 Hydraulic tool lifters. Unless otherwise specified (see 6.2.1), a hydraulic tool lifter mechanism shall be provided for both the crossrail toolhead and side toolhead. Means shall be provided for disengaging the hydraulic tool lifter mechanism.

* 3.4.8 Worktable. The worktable shall have the dimensions, quantity and size of T-slots, and number of rows and diameter of stop-pin holes specified in table I. The T-slots shall also conform to ANS B5.1. Unless otherwise specified (see 6.2.1), the worktable cutting and return speed and cutting force shall be not less than that shown as "standard" in table I. The table ways shall be automatically lubricated by a pressurized, filtered lubrication system capable of providing a film of oil over the entire sliding surface of the table ways in not less than 10 seconds after the table drive motor is started. Type II machine shall be provided with an automatic control that will disengage the table drive system in the event of lubrication failure. The table drive mechanism shall conform to 3.4.8.1, and shall be capable of smoothly accelerating, driving, decelerating, and reversing the table without shock, slamming, or sluggish action at the table reversal trip points when operating at any table drive speed specified in table I.

3.4.8.1 Worktable drive system. The worktable shall be driven by a hydraulic system consisting of an electric motor-driven hydraulic pump, one or more hydraulic cylinders, a fluid sump and all necessary valves, piping and pressure and flow controls. The hydraulic system shall conform to 3.4.10. The drive system shall provide an infinite number of table cutting and return speeds within the ranges specified in table I. The return speed controls shall be completely independent of the cutting speed control, making it possible to vary the return speed for any selected cutting speed. The system shall have adjustable trip stops at both ends of the table for adjustment of the table reversal points. To facilitate set-up and permit temporary short-stroking when the drive system is adjusted for long cuts, the machine shall have a manually

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operated reversing control that will allow the operator to manually interrupt the cut at any point within the adjusted stroke and instantly return the table without disturbing the adjustment of the trip stops. The table stroke length shall be as specified in 1.2, and shall be one of the stroke lengths selected from table I. The hydraulic pump pressure and capacity, and the diameter and stroke of the table drive cylinders shall be such that they are capable of providing the table stroke lengths, cutting speeds, return speeds and cutting forces specified herein.

3.4.8.1.1 Worktable drive system controls. The controls for regulating the cutting and return movements of the table, and the table "stop-start" control, shall be located in a position allowing close observation of the cutting tool action of the workpiece. A stop-start control shall be provided on both sides of the table to enable the operator to stop or start the table motion from either side.

3.4.8.2 Table safety stops. Means shall be provided to prevent the table from sliding off the bedways in the event of failure of any part of the table drive mechanism.

* 3.4.9 Electrical equipment. Unless otherwise specified (see 6.2.1), the electrical system shall conform to all requirements in this paragraph. The system shall conform to all requirements of ANSI/NFPA No. 79. An identified terminal shall be provided for grounding the machine when installed. The terminal shall be suitable for connection of the size grounding conductor specified in the ANSI standards. It shall be possible to operate all electrical equipment on the machine after connection to a single power source. Unless otherwise specified (see 6.2.1), the planer shall operate on both 230 volts and 460 volts, 3 phase, 60 hertz power and shall be initially wired for operation on 230 volts.

* 3.4.9.1 Motors. Motors shall have ball or roller bearings. Motors shall be 230/460 volt, 3 phase, 60 Hz, AC. Unless otherwise specified (see 6.2.1), each motor shall be single speed, and the enclosure shall meet the requirements for a dripproof enclosure and shall be connected for 230 volt operation. The main drive motor horsepower shall be not less than that shown "as standard" in table I.

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3.4.10 Hydraulic system. The hydraulic system shall be complete, and shall include all pumps,. valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided on the high pressure line from the hydraulic pump, and on the low side of each pressure reducing valve to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid. The hydraulic reservoir capacity shall be not less than the requirements of table I, and shall have means for determining fluid level.

* 3.5 Sizes and characteristics. Unless otherwise specified, (see 6.2.1) the size and stroke length of the machines shall meet the requirements of table I. If rated stroke length requirements are different than that in table I, they shall be specified in increments of 2 feet from 8 feet to 20 feet.

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* TABLE I - SIZES AND CHARACTERISTICS*

CHARACTERISTIC	Type I Short Stroke Type	Type II Medium Stroke Type		
Nominal Size, Table Width by Table Stroke	20 inch by 3' or 6' stroke	30 inch by 8', 14' and 20' stroke	36 inch by 8', 14' and 20' stroke	42 inch by 8', 14' and 20' stroke
Worktable				
Width, total, inches	20	30	36	42
Length, total, inches	Rated stroke length, plus 12 in.	<---- Not less than rated stroke length ---->		
T-slots, quantity x width in inches	3 x 13/16	5 x 13/16	5 x 13/16	5 x 13/16
Stop-pin holes, no. of rows x dia. in inches	4 x 1	4 x 1	6 x 1	6 x 1
Stroke length, feet	3' or 6', as specified see 6.2.1 (b)	<---- 8', 14' or 20' as specified see 6.2.1 (b) ---->		
Standard high speed				
Table cutting and return speed selections	Infinite	Infinite	Infinite	Infininte
Table cutting speed, ft/min	185	20 - 300	20 - 300	20 - 300
Table return speed ft/min	185	300	300	300
Table cutting force at max. cutting speed, lbs.	2800	6000	6000	6000
Standard low speed				
Table cutting and return speed selections		Infinite	Infinite	Infinite
Table cutting speed ft/min		80 - 100	80 - 100	80 - 100
Table return speed ft/min		300	300	300
Table cutting force at max cutting speed, lbs		14,400	14,400	14,400
Crossrail				
Vertical travel, max. distance above table surface, inches standard	24	24	24	24
optional (when specified, see 3.4.4 and 6.2.1)	32	30, 36 or 42	30, 36 or 42	30, 36 or 42
Crossrail toolhead				
Horizontal feed:	Manual and power	Manual and power	Manual and power	Manual and power
No. of power feed selections	Infinite	Infinite	Infinite	Infinite
Power feed rate, inches per stroke, range**	0.005 to 0.125	0.005 to 0.500	0.005 to 0.500	0.005 to 0.500
Total tool travel, distance from column, inches	30	32	38	44
Vertical feed:	Manual and power	Manual and power	Manual and power	Manual and power
No. of power feed selections	Infinite	Infinite	Infinite	Infinite
Power feed rate, inches per stroke, range**	0.005 to 0.0625	0.005 to 0.500	0.005 to 0.500	0.005 to 0.500
Total tool travel, inches	9	9	9	9
Tool shank size, inches	1-7/8 x 2-1/4	2-1/4 x 2-3/4	2-1/4 x 2-3/4	2-1/4 x 2-3/4
Siderail toolhead				
Horizontal feed:	Manual	Manual	Manual	Manual
Total tool travel, inches	9	12	12	12
Vertical feed:	Manual and power	Manual and power	Manual and power	Manual and power
No. of power feed selections	Infinite	Infinite	Infinite	Infinite
Power feed rate, inches per stroke, range **	0.005 to 0.125	0.005 to 0.500	0.005 to 0.500	0.005 to 0.500
Total travel above plane of table surface, in.				
Machines with "standard" crossrail vert. travel	18	17	17	17
Machines with "optional" crossrail vert. travel	26	23, 29 or 35	23, 29 or 35	23, 29 or 35
Main drive motor horsepower				
Standard	15	50	50	50
Hydraulic system capacity, gallons	50 (3' stroke), 110 (6' stroke)	52 (8' stroke), 58 (14' stroke), 64 (20' stroke)		

* All numerical values are "not less than"
** All ranges are from "not more than the minumum to "not less than the maximum".

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3.6. Performance. Each machine shall be capable of producing cutting forces at maximum table cutting speed in accordance with table I.

3.6.1 Alignment tolerances. Each machine shall conform to the alignment tolerances specified in table II.

3.7 Standard equipment. Unless otherwise specified (see 6.2.1), one set of hand tools, normally furnished by the manufacturer, shall be furnished.

3.8 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2.1).

3.9 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

* 3.10 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a lubrication chart or plate shall be permanently and securely attached to each machine. If a chart is furnished, it shall be placed in a transparent plastic folder, or permanently sealed between clear plastic sheets, with suitable means for mounting. The following information shall be furnished on the chart or plate:

Points of application

Service interval

Type of lubricant

Viscosity

* 3.11 Nameplate. Unless otherwise specified (see 6.2.1), a corrosion resistant metal nameplate shall be securely attached to each machine. The nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix keyed to the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided that the entry for each caption is clear as to its identity.

Nomenclature

Manufacturer's name

Manufacturer's model designation

Manufacturer's serial number

Table width, maximum stroke length

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TABLE II

ALIGNMENT TOLERANCES

REQUIREMENTS	TOLERANCES
Worktable	
Table surface level in longitudinal and transverse directions	0.0005 inch per foot
Table surface flat in longitudinal and transverse directions	+0.0005 inch per 3 feet
Rise and fall of table when traversed full stroke (measured on inboard and outboard edges of table)	0.0005 inch TIR per foot
Table T-slots parallel to table travel	0.0005 inch TIR per 3 feet
Crossrail Toolhead	
Toolhead horizontal travel parallel to table top surface	0.0007 inch TIR per foot
Toolhead downfeed square with table top surface (with toolhead adjusted to 0 degrees swivel angle with aid of 4 power magnifier)	0.001 inch TIR per foot
Siderail Toolhead	
Toolhead vertical travel square with table top surface	0.0007 inch TIR per foot
Toolhead infeed parallel to table to p surface (with toolhead adjusted to 0 degree swivel angle with aid of 4 power magnifier)	0.001 inch TIR per foot

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Power input characteristics and rating

Date of manufacture

Contract Number or Order Number

National Stock Number or Plant Equipment Code

3.12 Technical data. Technical data shall be furnished as specified (see 6.2.2).

3.13 Workmanship. Workmanship of the machines and accessories shall be commensurate with the requirements specified herein, and shall be of a quality prevailing among manufacturers normally producing equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the first article when required in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and the tests in 4.6. Failure of the item to pass any examination,test or inspection shall be cause for rejection.

* 4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.5, the test in 4.6.1, and the inspection in 4.6.4. Failure of the machine to

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pass the examination, test, or inspection shall be cause for rejection.

4.5 Examination. The machine and equipment shall be examined to determine compliance with the requirements of 3.2 through 3.5 and 3.7 through 3.13.

4.6 Tests.

4.6.1 Operation test. The machine shall be operated at no load for not less than one hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

* 4.6.2 Performance test. The machine shall be subjected to the following cutting force test: A calibrated pressure gauge shall be installed in the hydraulic system to indicate the hydraulic pressure at the inlet of the hydraulic cylinder that operates the worktable. A workpiece of suitable size shall be installed on the table, a suitable cutting tool shall be installed in the railhead tool holder, and the table speed shall be adjusted to the maximum cutting speed shown in table I. A cut shall be started, and the cutting depth and feed shall be increased until the table speed starts to fall below the maximum cutting speed. The depth of cut or feed shall then be decreased until the table is again moving at its rated speed at which time the pressure gauge reading shall be taken when the table is in the middle of its working stroke. The table maximum cutting force shall be computed using the following equation:

Table Cutting Force = (.80) x (maximum pressure gauge reading) x (total working area of hydraulic piston as calculated from certified manufacturing drawings)

The cutting force at maximum table speed shall conform to the requirments of table I. During this test there shall be no evidence of hydraulic fluid leakage, overheating of motors, or permanent distortion of any machine component or structural members.

4.6.3 Alignment accuracy test. The machine shall be tested to determine that it complies with the alignment tolerance requirements of table II. The instruments and methods used shall be at the option of the contractor, provided they accurately determine the alignment tolerance of the machine.

* 4.6.4 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of Section 5.

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* 5.1 Preservation, packing, and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall conform to the requirements of MIL-STD-1188.

6. NOTES

6.1 Intended use. The planers covered by this specification are intended for rough and finish planing of metal workpieces.

* 6.2 Ordering data.

* 6.2.1 Procurement requirements. Purchasers should excercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type and size required (see 1.2).

c. Specification and standards, if different (see 2.1.1).

d. When first article is required for inspection and approval (see 3.1 and 4.3).

e. If machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.1).

f. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI) state required system (3.2.1.1).

g. Dual calibrations (US and SI); if required (see 3.2.1.2).

h. Specify additional safety and health requirements (see 3.2.4).

i. Painting, if different (see 3.3.5).

j. If table ways are required to be a specific type, state required type (see 3.4.3).

k. Crossrail vertical travel, if different, from dimension of table I (see 3.4.4).

l. Crossrail elevating mechanism, if different (see 3.4.4).

m. Crossrail toolhead, if different (see 3.4.4.1).

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n. Type I machine crossrail toolhead rapid traverse mechanism, if required (see 3.4.4.1.2).

o. Siderail toolhead, if different (see 3.4.5).

p. Type I machine sidehead rapid elevation system, if required (see 3.4.5.2).

q. Hydraulic tool lifters, if different (see 3.4.7).

r. Worktable cutting and return speed and cutting force, if different from "standard" in table I (see 3.4.8).

s. Electrical equipment, if different (see 3.4.9).

t. Motor characteristics, if different (see 3.4.9.1).

u. Size and stroke length, if different (see 3.5).

v. Standard equipment, if different (see 3.7).

w. Optional equipment, if required (see 3.8.).

x. List of repair parts required (see 3.9).

y. Lubrication chart, if different (see 3.10).

z. Nameplate, if different (see 3.11).

aa. First Article inspection, if different (see 4.3).

bb. Quality conformance inspection, if different (see 4.4).

cc. Level of packaging and marking required (see 5.1).

6.2.2 Contract data requirements. Required technical data such as operator's manuals, parts lists, foundation and anchor bolt plans, wiring diagrams and other instructions for operation and maintenance as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated into the contract.

* 6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Armed Services Procurement Regulation. The first article should be a reproduction sample, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examinations, test and approval of the first article.

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* 6.4 Safety and health requirements. Paragraph 3.2.4 requires compliance only with those OSHA requirements that concern the machine itself. It does not require compliance with those OSHA requirements that concern "the machine in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, heat, etc. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual machines in the environment), the procuring agency is advised to analyze the existing hazard level in the proposed operating environment, and specify additional machine requirements that will integrate the new machine into its future operating environment. If specific point-of- operation guarding is required, the procuring agency should specify the exact configuration of the guard required as in most cases the guard configuration is dependent on the size and configuration of the workpieces that will be machined. The above, and any other additional safety and health requirements should be specified in detail under 6.2.1(h).

* 6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, delections) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements in this document based on the entire content irrespective of the marginal notations and relations to the previous issue.

* 6.6 Cross-reference of classification changes. Machine classification changes from the previous issue of this specification are as follows:

MIL-P-30136A	MIL-P-30136B
Type I Size 20	Type I Size 1
	Type I Size 2
Type II Size 30	Type II Size 1
36	Type II Size 2
42	Type II Size 3
	Type II Size 4
	Type II Size 5
	Type II Size 6
	Type II Size 7
	Type II Size 8
	Type II Size 9

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Navy - SH
Air Force - 99

Preparing Activity:
DLA - IP

Project Number:
3418-0025

Review Activities:
DLA-GS

User Activities:
Navy - MC
Army - AV

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-P-80136B, Planers, Metal Cutting, Openside, Hydraulic, Shaper-Planer Type

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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